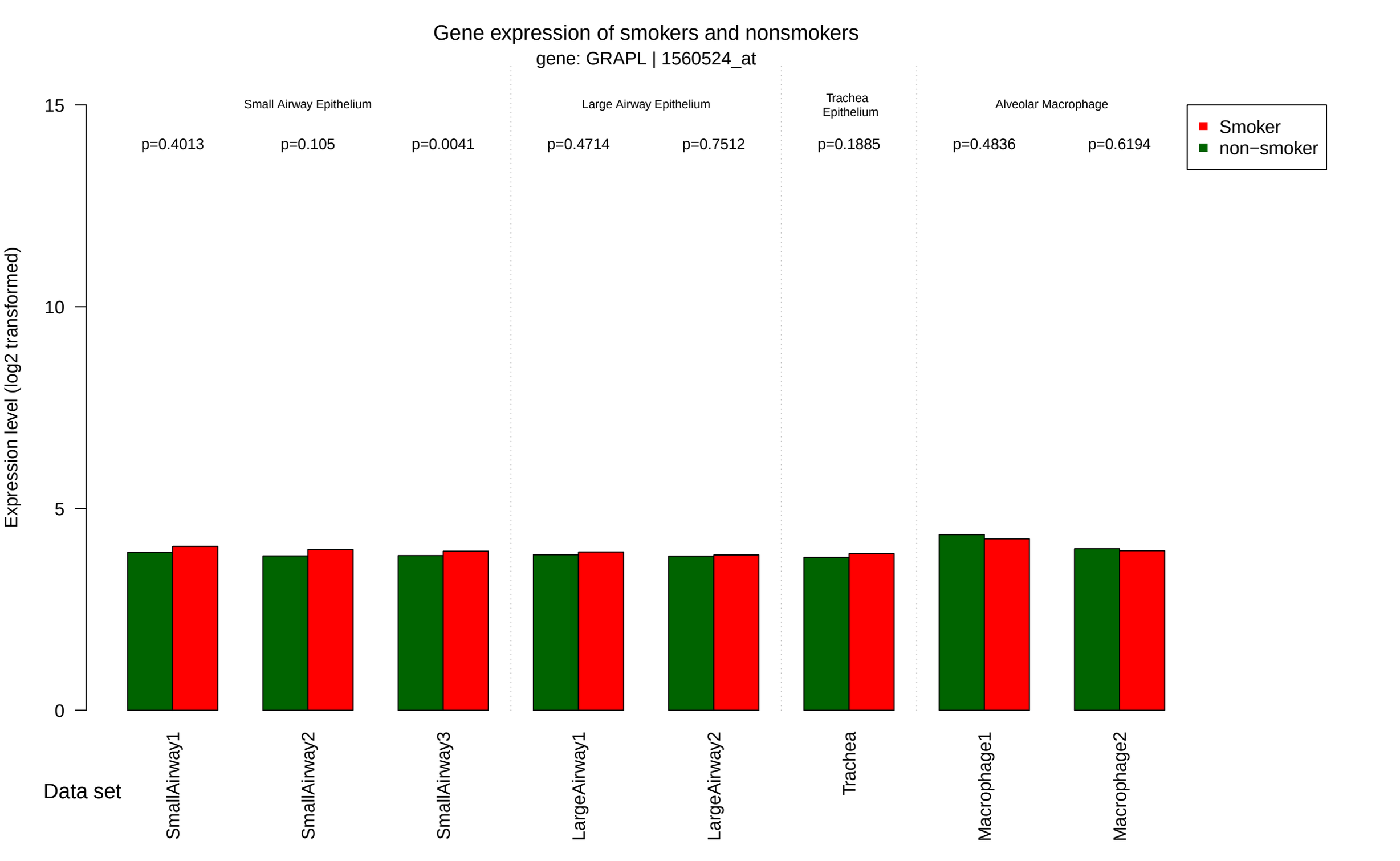
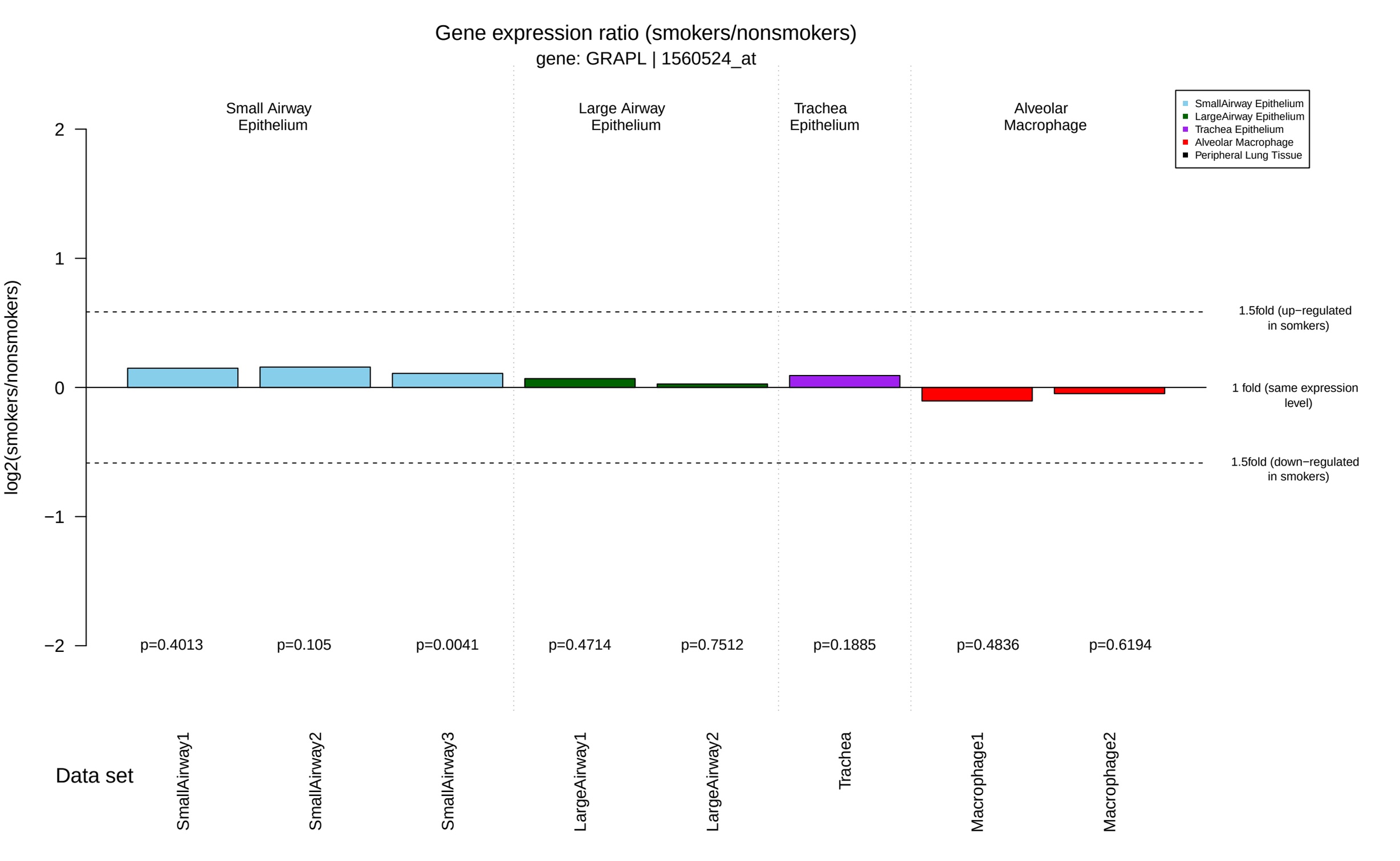
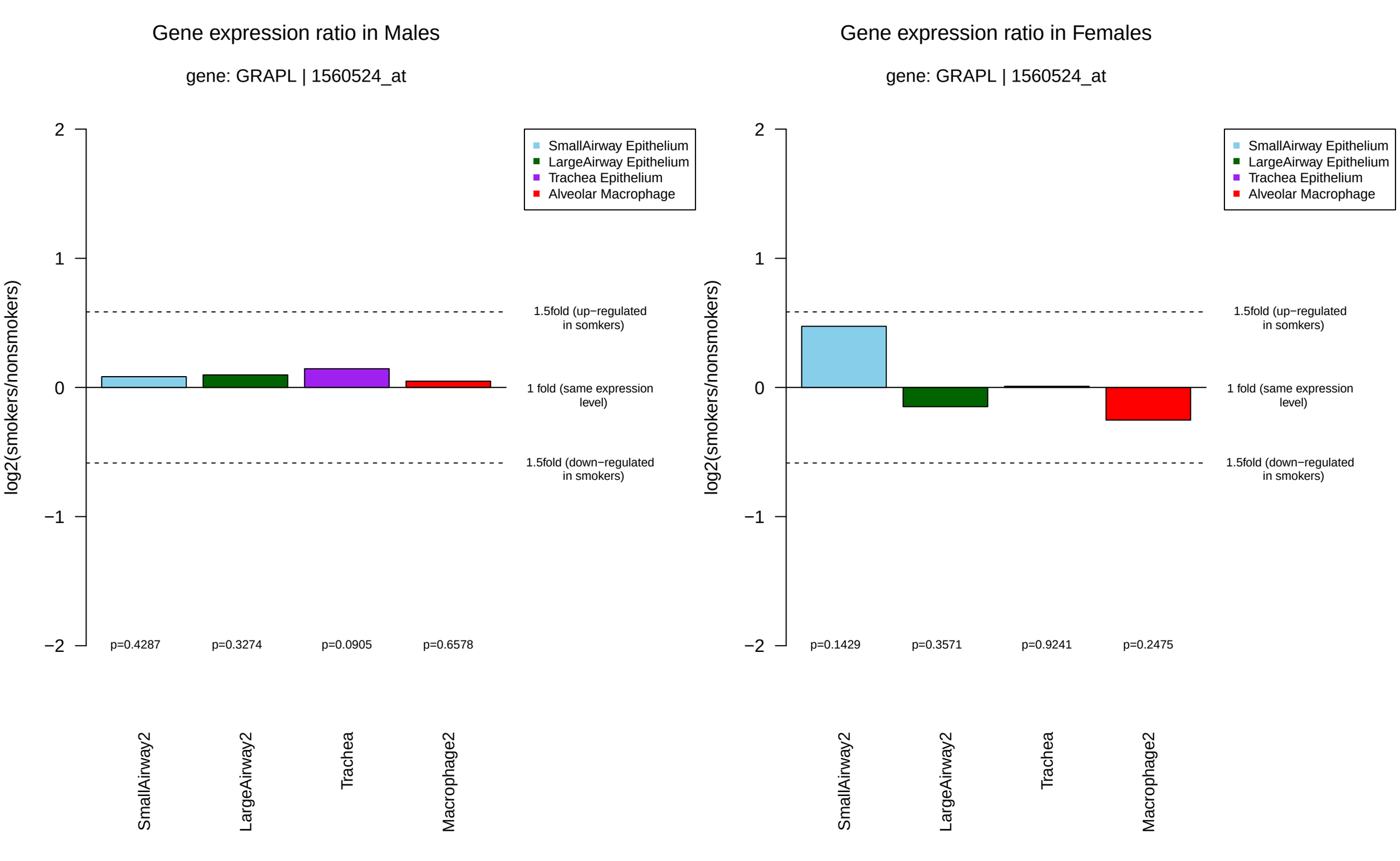
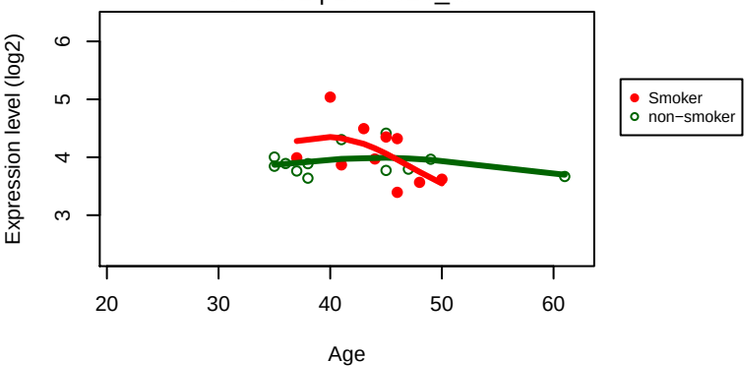
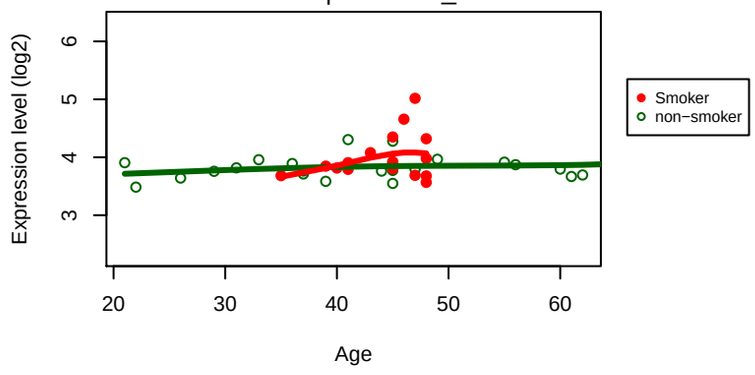
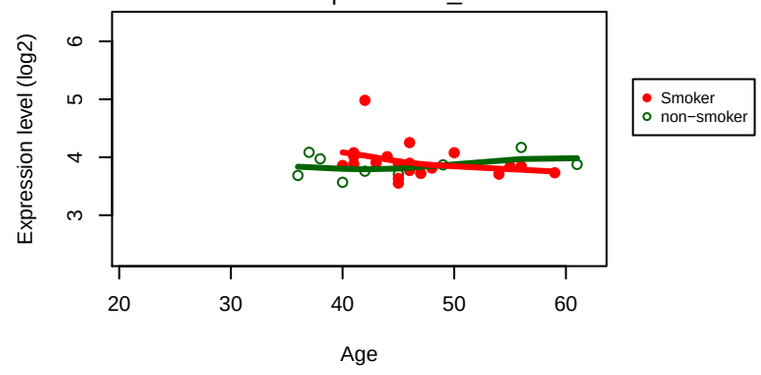

Gene expression of smokers and nonsmokers
gene: GRAPL | 1560524_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4013
p=0.105
p=0.0041
p=0.4714
p=0.7512
p=0.1885
p=0.4836
p=0.6194
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

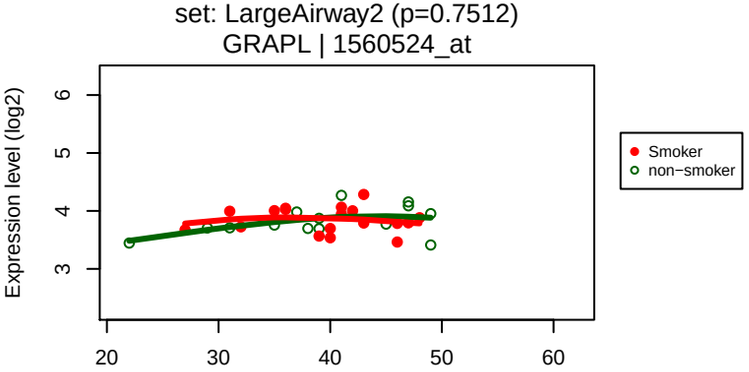

Gene expression ratio (smokers/nonsmokers)
gene: GRAPL | 1560524_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4013
p=0.105
p=0.0041
p=0.4714
p=0.7512
p=0.1885
p=0.4836
p=0.6194
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

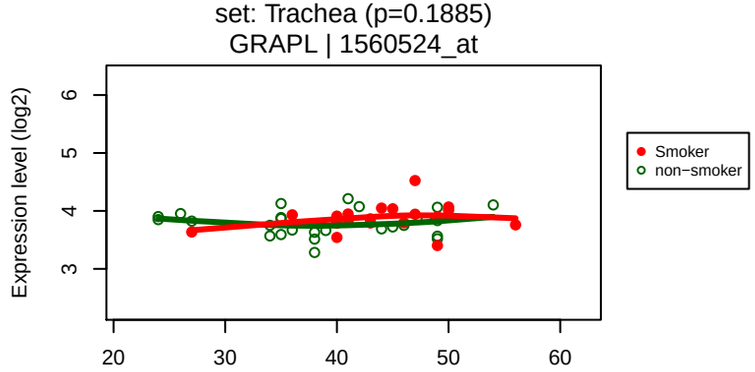

Gene expression ratio in Males
gene: GRAPL | 1560524_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4287
p=0.3274
p=0.0905
p=0.6578
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRAPL | 1560524_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1429
p=0.3571
p=0.9241
p=0.2475
SmallAirway2
LargeAirway2
Trachea
Macrophage2

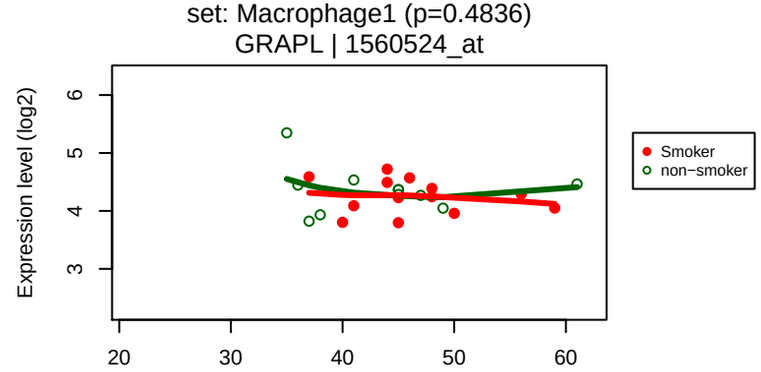

set: SmallAirway1 (p=0.4013)
GRAPL | 1560524_at
Expression level (log2)
Age
Smoker
non−smoker

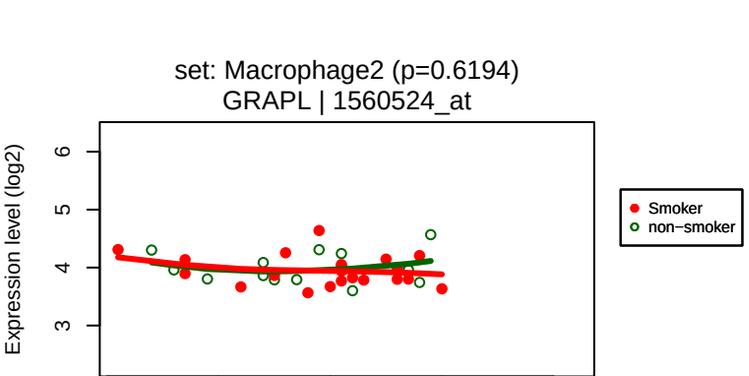

set: SmallAirway2 (p=0.105)
GRAPL | 1560524_at
Expression level (log2)
Age
Smoker
non−smoker

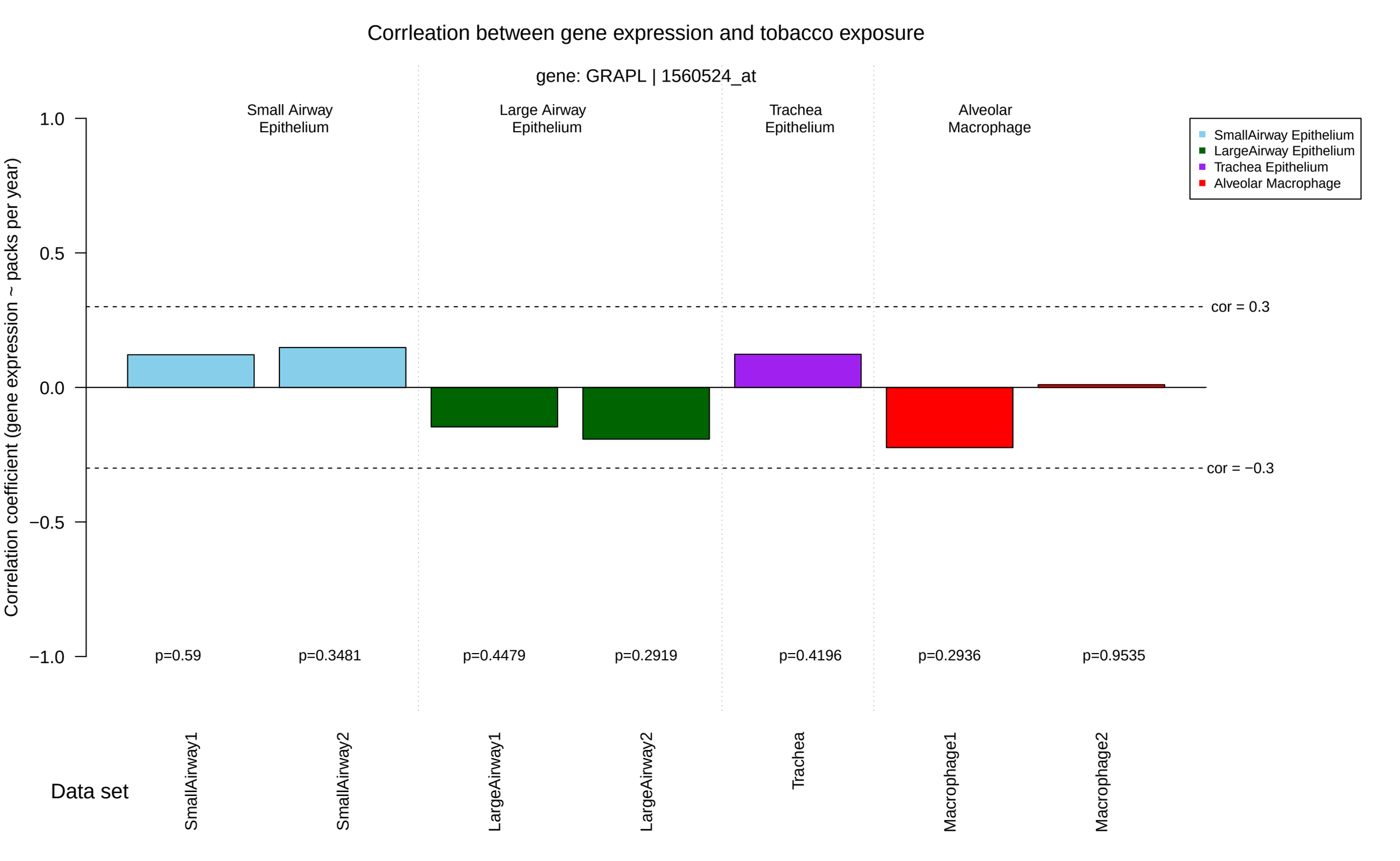

set: LargeAirway1 (p=0.4714)
GRAPL | 1560524_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7512)
GRAPL | 1560524_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1885)
GRAPL | 1560524_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4836)
GRAPL | 1560524_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6194)
GRAPL | 1560524_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRAPL | 1560524_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.59
p=0.3481
p=0.4479
p=0.2919
p=0.4196
p=0.2936
p=0.9535
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2